

The Atomic Energy Network (ænet) (release 2.0.5)

Nongnuch Artrith

September 25, 2026

Contents

1	What is ænet?	3
2	License	3
3	Installation	4
3.1	Short installation summary	4
3.2	Detailed installation instructions	5
3.2.1	Compilation of external libraries that are distributed with ænet	6
3.2.2	Build ænet	6
3.2.3	Set up the Python interface	7
4	Installing a binary release	7
4.1	Download the correct archive	7
4.2	Verify and extract the download	8
4.3	Run the executables	9
4.4	Small generate example	9
4.5	Troubleshooting	11

5	Version queries and release versions	11
6	General concepts	12
7	References	14
8	ANN potential construction	14
8.1	Structural energy reference data	15
8.1.1	Example <code>ænet</code> XSF file of an isolated structure	15
8.1.2	Example <code>ænet</code> XSF file of a periodic structure	16
8.2	Invariant basis (structural fingerprint)	16
8.2.1	List of keywords	17
8.2.2	Input file template using a pre-defined basis set (<code>atomtype.stp</code>)	17
8.2.3	Input file template using explicit basis function definitions (<code>atomtype.stp</code>)	18
8.2.4	Input file example using a Chebyshev basis set (<code>Ti.fingerprint.stp</code>)	18
8.2.5	Input file example using explicit Behler2011 basis functions (<code>Ti.fingerprint.stp</code>)	19
8.3	Training set generation with <code>generate.x</code>	21
8.3.1	Alphabetic list of keywords	22
8.3.2	Input file template (<code>generate.in</code>)	22
8.3.3	Input file example (<code>generate.in</code>) for TiO_2	23
8.4	ANN potential training with <code>train.x</code>	24
8.4.1	Alphabetic list of keywords	25
8.4.2	Training methods	26
8.4.3	Input file template (<code>train.in</code>)	27
8.4.4	Example input file (<code>train.in</code>)	28
8.5	Restarting training from existing ANN potential	29
9	Using ANN potentials for atomistic simulations	29

9.1	Prediction of structural energies and atomic forces with <code>predict.x</code>	29
9.1.1	Alphabetic list of keywords	30
9.1.2	Input file template (<code>predict.in</code>)	31
9.1.3	Input file example (<code>predict.in</code>) for TiO_2	31
9.2	ASE Interface: <code>aenet-predict.py</code> and <code>aenet-md.py</code>	32
9.2.1	Alphabetic list of keywords	33
9.2.2	Input file template (<code>input.json</code>)	33
9.2.3	Input file example (<code>input.json</code>)	33
10	Acknowledgment	34
11	Questions?	34
12	Bibliography	34

1 What is `aenet`?

The Atomic Energy NETWORK (`aenet`) package is a collection of tools for the construction and application of atomic interaction potentials based on artificial neural networks (ANN). The `aenet` code allows the accurate interpolation of structural energies, e.g., from electronic structure calculations, using ANNs. ANN potentials generated with `aenet` can then be used in larger scale atomistic simulations and in situations where extensive sampling is required, e.g., in molecular dynamics or Monte-Carlo simulations.

2 License

Copyright (C) 2012-2026 The Aenet Developers. See `DEVELOPERS` for the developer list.

The `aenet` source code is subject to the terms of the Mozilla Public License, v. 2.0. If a copy of the MPL was not distributed with this file, You can obtain one at <http://mozilla.org/MPL/2.0/>.

This program is distributed in the hope that it will be useful, but WITHOUT

ANY WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the Mozilla Public License, v. 2.0, for more details.

3 Installation

3.1 Short installation summary

1. Compile the L-BFGS-B library

- Enter the directory `./lib`
`$ cd ./lib`
- Adjust the compiler settings in the "Makefile"
- Compile the library with
`$ make`

The library file `liblbfgsb.a`, required for compiling **ænet**, will be created.

2. Compile the **ænet** package

- Enter the directory `./src`
`$ cd ./src`
- Compile the **ænet** source code with
`$ make -f makefiles/Makefile.XXX`
where `Makefile.XXX` is an appropriate Makefile.
To see a list of available Makefiles just type:
`$ make`

The following executables will be generated in `./bin`:

- **generate.x**: generate training sets from atomic structure files
- **train.x**: train new neural network potentials
- **predict.x**: use existing ANN potentials for energy/force prediction

3. (Optional) Install the Python interface

- Enter the directory `./python`
`$ cd ./python`
- Install the Python module with
`$ python setup.py install --user`

This will set up the Python **ænet** module for the current user, and it will also install the user scripts **ænet-predict.py** and **ænet-md.py**.

3.2 Detailed installation instructions

Except for a number of Python scripts, **ænet** is developed in Fortran 95/2003. Generally, the source code is tested with the free GNU Fortran compiler and the commercial Intel Fortran compiler, and the Makefile settings for these two compilers are provided. While the **ænet** source code should be platform independent, we mainly target Linux and Unix clusters and **ænet** has not been tested on other operating systems.

ænet requires three external libraries:

1. BLAS (Basic Linear Algebra Subprograms),
2. LAPACK (Linear Algebra PACKage),
3. And the L-BFGS-B optimization routines by Nocedal et al.

Usually, some implementation of BLAS and LAPACK comes with the operating system or the compiler. If that is not the case, the libraries can be obtained from [Netlib.org](http://netlib.org). `libblas.a` and `liblapack.a` have to be in the system library path in order to compile **ænet**.

The L-BFGS-B routines, an implementation of the bounded limited-memory Broyden-Fletcher-Goldfarb-Shanno algorithm, is distributed on the [homepage of the authors](#) (Nocedal et al.). For the user's convenience we have decided to distribute the original L-BFGS-B files along with **ænet** package, so you do not have to actually download the library yourself. However, each application of **ænet** should also acknowledge the use of the L-BFGS-B library by citing: R. H. Byrd, P. Lu and J. Nocedal, *SIAM J. Sci. Stat. Comp.* **16** (1995) 1190-1208.

ænet's Python interface further relies on [NumPy](#) and on the [Atomic simulation Environment](#), so these dependencies have to be available when the **ænet** Python module is set up.

3.2.1 Compilation of external libraries that are distributed with **ænet**

All external libraries needed by the **ænet** code are in the directory `./lib`. Currently, only one external library is distributed with **ænet**, the L-BFGS-B library (see above).

To compile the external libraries

1. Enter the directory `./lib`

```
$ cd ./lib
```

2. Adjust the compiler settings in the "Makefile"

The Makefile contains settings for the GNU Fortran compiler (**gfortran**) and the Intel Fortran compiler (**ifort**). Uncomment the section that is appropriate for your system.

3. Compile the library with

```
$ make
```

The static library `liblbfgsb.a`, required to build **ænet**, will be created.

3.2.2 Build **ænet**

The **ænet** source code is located in `./src`.

1. Enter `./src`

```
$ cd ./src
```

2. To see a short explanation of the Makefiles that come with **ænet**, just run **make** without any options.

```
$ make
```

Select the Makefile that is appropriate for your computer.

3. Compile with

```
$ make -f makefiles/Makefile.XXX
```

where **Makefile.XXX** is the selected Makefile.

Three executables will be generated and stored in `"/bin"`:

- `generate.x`: generate training sets from atomic structure files
- `train.x`: train new neural network potentials
- `predict.x`: use existing ANN potentials for energy/force prediction

3.2.3 Set up the Python interface

1. Enter the directory `"/python"`

```
$ cd ./python
```
2. Install the Python module with

```
$ python setup.py install --user
```

This will set up the Python `ænet` module for the current user, and it will also install the user scripts `ænet-predict.py` and `ænet-md.py`.

4 Installing a binary release

Binary releases provide the serial GNU Fortran backend without requiring a Fortran compiler. The first supported platforms are:

- macOS 15 or newer on Apple silicon (`arm64`), using system Accelerate;
- Linux x86_64 with glibc 2.35 or newer, as provided by Ubuntu 22.04.

Intel processors, Linux arm64, older operating systems, MPI executables, and Intel Fortran builds are not included in the first binary release. Source builds remain available for other supported configurations.

4.1 Download the correct archive

Open the [ænet releases page](#) and select the required archive and its matching `.sha256` file:

- `ænet-<version>-macos-arm64-gnu-serial.tar.gz`

- `aenet-<version>-linux-x86_64-gnu-serial.tar.gz`

Here, `<version>` is the release version without the leading `v` used by the Git tag. For example, tag `v2.1.0` would contain files beginning with `aenet-2.1.0-`. The direct-download URL convention is:

`https://github.com/atomisticnet/aenet/releases/download/v<version>/<archive>`

The first public binary release has not yet been published. The final links and browser-download behavior will be verified as part of that release; use the release page rather than guessing a version.

4.2 Verify and extract the download

Keep the archive and checksum sidecar in the same directory. On macOS, verify and extract them with:

```
shasum -a 256 -c aenet-<version>-macos-arm64-gnu-serial.tar.gz.sha256
xattr -d com.apple.quarantine aenet-<version>-macos-arm64-gnu-serial.tar.gz
tar -xzf aenet-<version>-macos-arm64-gnu-serial.tar.gz
```

The `xattr` command is the explicit approval step for a quarantined browser download. Run it only after the checksum reports OK and the files came from the official release. If the archive has no quarantine attribute, `xattr` reports `No such xattr` and can be omitted.

On Linux, use:

```
sha256sum -c aenet-<version>-linux-x86_64-gnu-serial.tar.gz.sha256
tar -xzf aenet-<version>-linux-x86_64-gnu-serial.tar.gz
```

A successful checksum command reports OK. Do not use an archive whose checksum fails. Each archive extracts into a directory with the same name as the archive without `.tar.gz`. This directory contains `bin/`, `tools/`, `lib/`, `include/`, `licenses/`, `metadata.json`, and `manifest.txt`.

4.3 Run the executables

The commands can be run from the extracted directory without setting a library search path. For example:

```
cd aenet-<version>-<platform>-gnu-serial
./bin/generate.x --version
./bin/train.x --version
./bin/predict.x --version
```

Each command prints its program name and the release version. To make the main programs and auxiliary tools available in the current shell, run:

```
AENET_ROOT="$PWD"
export PATH="$AENET_ROOT/bin:$AENET_ROOT/tools:$PATH"
```

Add equivalent commands to the shell startup file only after choosing a permanent installation directory. The executables locate their bundled GNU runtime libraries through relative paths, so `DYLD_LIBRARY_PATH` and `LD_LIBRARY_PATH` should not be set for a normal binary installation. Developers compiling against the C interface can find `aenet.h` under `include/` and the shared and static libraries under `lib/`.

4.4 Small generate example

The following self-contained example creates a two-atom copper structure and converts it into a training-set file. Run it in a new writable directory after adding the extracted `bin/` directory to `PATH` as shown above.

```
mkdir aenet-binary-example
cd aenet-binary-example
```

Create the descriptor setup:

```
cat > Cu.stp <<'EOF'
DESCR
Binary installation example
END DESCR
```

```

ATOM Cu
ENV 1
Cu
RMIN 0.5
BASIS type=chebyshev
radial_Rc=5 radial_N=3 angular_Rc=5 angular_N=2
EOF

```

Create the two-atom structure:

```

cat > structure.xsf <<'EOF'
# total energy = -1.0 eV
ATOMS
Cu 0.0 0.0 0.0 0.0 0.0 0.0
Cu 2.5 0.0 0.0 0.0 0.0 0.0
EOF

```

Create the `generate.x` input and run it:

```

cat > generate.in <<'EOF'
OUTPUT example.train
TYPES
1
Cu 0.0
SETUPS
Cu Cu.stp
FILES
1
structure.xsf
EOF

```

```

generate.x generate.in
test -s example.train

```

The final command succeeds when `generate.x` creates a nonempty `example.train` file. The synthetic structure only tests the installed executable; it is not scientific training data.

4.5 Troubleshooting

- **Architecture errors.** For cannot execute binary file, bad CPU type, or a similar error, check `uname -m`. The macOS archive requires `arm64` and the Linux archive requires `x86_64`. Download the archive matching both the operating system and architecture.
- **Operating-system version or GLIBC errors.** The supported minimum is macOS 15 on Apple silicon or glibc 2.35 on Linux `x86_64`. Upgrade the operating system or build `ænet` from source on the older system.
- **Permission denied.** Extract with `tar` rather than copying individual files from the archive. The manifest records executable permissions; rerun the checksum and extract a fresh archive if they were lost.
- **Missing shared libraries.** Do not move individual executables or libraries out of the extracted tree. Keep `bin/`, `tools/`, and `lib/` together, and remove custom `DYLD_LIBRARY_PATH` or `LD_LIBRARY_PATH` settings before retrying.
- **Checksum failures.** Download both files again from the same GitHub release. Do not combine an archive and sidecar from different versions.
- **macOS cannot open or verify the application.** The binaries have valid ad-hoc signatures but are not Developer ID signed or notarized. Delete the extracted directory, verify the downloaded archive again, remove its quarantine attribute with the documented `xattr` command, and extract it again. Do not disable system-wide security settings. Signing and notarization are tracked as a future usability improvement.

When reporting a problem, include the archive name, output of `uname -a`, the failing command and complete error, and `metadata.json` from the extracted directory.

5 Version queries and release versions

`src/VERSION` is the canonical release version, in `MAJOR.MINOR.PATCH` form (currently `2.0.5`). Git release tags add a `v` prefix, for example `v2.0.5`. CMake reads this file and automatically reconfigures on changes.

Each main executable accepts a standalone `--version` argument. For example, `train.x_serial --version` prints exactly `train.x 2.0.5` followed by a newline and exits successfully. The reported program name omits build suffixes. This query needs no input files and creates no application output files. MPI builds print once and finalize all ranks. Missing arguments or nonexistent input files produce a nonzero exit.

CMake shared libraries use the full release as `VERSION` and its major component as `SOVERSION`. An incompatible public ABI change requires a major release. This introduces a versioned library identity; downstream binaries should be relinked when adopting this convention. Static library names and existing executable suffixes are unchanged. Legacy Makefiles also read `src/VERSION` for executable queries and retain their existing library naming conventions.

From `src/`, run `./prepare-release.sh MAJOR.MINOR.PATCH` to update the version and source license headers. The script rejects prefixed or prerelease versions before editing files. It prints the corresponding `vMAJOR.MINOR.PATCH` tag commands but does not commit, tag, or publish.

6 General concepts

ænet provides tools for the construction and application of artificial neural network (ANN) potentials. Users who just want to use **ænet** for simulations based on existing ANN potentials can safely skip over section [\[\[construction\]](#) that explains the construction of ANN potentials directly to section [9](#).

Potential construction using **ænet** is broken down into two separate tasks: (i) the compilation of reference structures and energies into a single training set file using the tool `generate.x` and (ii) the actual fit of the ANN potentials using the tool `train.x`. The usage of these tools is described in section [8](#).

Simulations based on existing ANN potentials is enabled by the `ænetLib` library. `ænetLib` provides routines for parsing ANN potential files and for energy and force evaluation. Part of the **ænet** package are sample implementations in Fortran and Python that interface with `ænetLib`. These tools are discussed in section [9](#).

A schematic of the interplay of the different **ænet** tools is shown in figure [1](#) (taken from reference [\[1\]](#)).

The **ænet** tools `generate.x`, `train.x`, and `predict.x` are controlled via

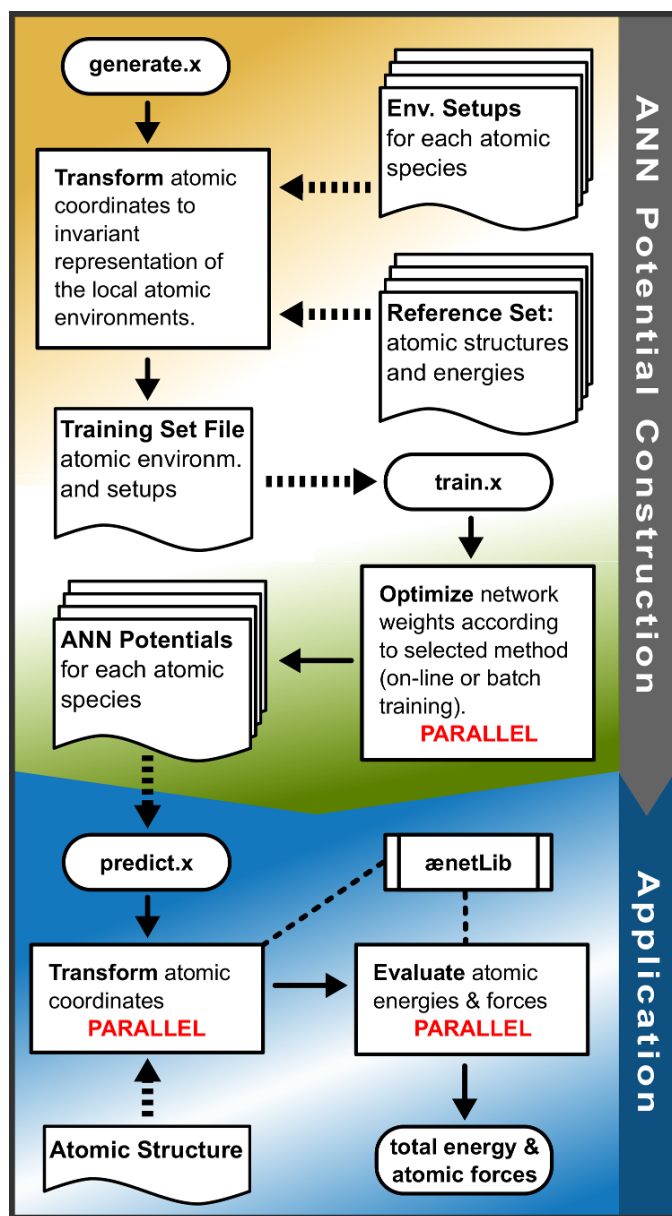


Figure 1: Schematic of the connection and workflow between the **ænet** tools (see reference [1]).

keyword-based input files. The keywords understood by each of the tools are discussed in their corresponding section; the order in which keywords appear in the input files is arbitrary. Keywords are not case sensitive.

7 References

Every scientific publication containing results that were produced with **ænet** should cite the appropriate original references.

The reference for the **ænet** package itself is: [1] N. Artrith and A. Urban, *Comput. Mater. Sci.* **114** (2016) 135-150.

If the local structural environment is represented by a *Chebyshev descriptor*, please cite: [2] N. Artrith, A. Urban, and G. Ceder, *Phys. Rev. B* **96** (2017) 014112.

The interpolation of *atomic* energies with ANNs was first published in: [3] J. Behler and M. Parrinello, *Phys. Rev. Lett.* **98** (2007) 146401.

If the local structural environment is represented by *symmetry functions*, please cite: [4] J. Behler, *J. Chem. Phys.* **134** (2011) 074106.

If the SOAP (*smooth overlap of atomic positions*) descriptor is used for the representation of the local structural environment, please cite: [5] A. P. Bartók, M. C. Payne, R. Kondor, and G. Csányi, *Phys. Rev. Lett.* **104** (2010) 136403.

The L-BFGS-B method is provided by a third party library. Whenever the method is used for training, please cite: [6] R. H. Byrd, P. Lu and J. Nocedal, *SIAM J. Sci. Stat. Comp.* **16** (1995) 1190-1208.

The references for the Levenberg-Marquardt method are: [7] K. Levenberg, *Q. Appl. Math.* **2** (1944) 164–168; [8] D. W. Marquardt, *SIAM J. Appl. Math.* **11** (1963) 431–441.

8 ANN potential construction

The construction of a new ANN potential is accomplished by interpolation of structural energies in a reference data set. The structure format used by **ænet** is explained in section 8.1.

To be useful for general atomistic simulations, ANN potentials have to be

invariant with respect to rotation/translation of the structure and exchange of equivalent atoms. Hence, the atomic coordinates have to be represented in a basis that fulfills these conditions. The specification of basis setups (*structural fingerprint* setups) is topic of section 8.2.

The transformation from Cartesian coordinates to invariant coordinates is the purpose of the tool `generate.x`, which iterates through a list of reference structures and transforms each structure’s coordinates using the method specified in the input file. The input file format for `generate.x` is discussed in section 8.3.

Finally, `train.x` implements different optimization algorithms that can be used for the training of ANN potentials. See section 8.4 for the usage of `train.x` and its input file format.

8.1 Structural energy reference data

The atomic structure format used by `ænet` for this purpose is a subset of the *XCrySDen Structure Format* (XSF) defined on the [XCrySDen homepage](#). Only the atomic positions of single isolated and periodic structures are parsed by `ænet`, i.e., `ænet` does neither support animated XSF files (trajectories) nor scalar fields (volumetric data). Additionally, `ænet` expects atomic symbols as type specifier, atomic numbers are currently not supported. The structural energy is included in the XSF file as a comment of the form `# total energy = XXX`, where `XXX` is the energy value. This has the advantage that the resulting file is still a valid XSF file and can be visualized with XCrySDen and various other visualization programs, such as [VMD](#) and [VESTA](#).

8.1.1 Example `ænet` XSF file of an isolated structure

The following is an example XSF file of an isolated (non-periodic) structure. Each line following the keyword `ATOMS` contains the atomic symbol, the three Cartesian coordinates, and the three components of the Cartesian force vector. In principle, any unit system may be used, but the length, energy, and force units have to be consistent. The example below uses Å, eV, and eV/Å.

Note that it is advisable to work with a greater number of decimals for the coordinates and atomic forces than used in the example to avoid loss of accuracy.

```
# total energy = -19543.67017695 eV
```

```

ATOMS
O   5.900  3.922  0.851 -0.001  0.001 -0.001
C   5.133  4.445  0.095  0.082  0.104  0.206
O   4.104  5.151  0.087  0.003 -0.001  0.000

```

8.1.2 Example `ænet` XSF file of a periodic structure

The following is an example of an XSF file of a periodic structure. The PRIMVEC block contains the lattice vectors in rows. For periodic structures, the number of atoms in the simulation cell has to be specified on the line following the keyword PRIMCOORD (the example is for 6 atoms). Note that the number 1 following the atom count is not relevant for `ænet`. The same comments as for the isolated structure example above apply.

```

# total energy = -4990.44928342 eV

CRYSTAL
PRIMVEC
    2.967  0.000  0.000
    0.000  4.648  0.000
    0.000 -0.000  4.648
PRIMCOORD
6 1
Ti 1.483  2.324  2.324  0.000  0.000  0.000
Ti 0.000  0.000  0.000  0.000  0.000  0.000
O  1.483  0.905  0.905  0.000 -0.004 -0.004
O  1.483  3.742  3.742  0.000  0.004  0.004
O  0.000  1.418  3.230  0.000  0.004 -0.004
O  0.000  3.230  1.418  0.000 -0.004  0.004

```

8.2 Invariant basis (structural fingerprint)

Currently, `ænet` implements two descriptors for the local atomic environment: the Artrith-Urban-Ceder descriptor based on a Chebyshev expansion [2] and the invariant *symmetry function* basis by Behler and Parrinello [3,4]. The code is designed such that implementing further methods is straightforward.

8.2.1 List of keywords

All keywords are case insensitive, but currently have to occur in the given order. Blank lines and lines starting with `!`, `#`, or `%` are ignored.

descr (optional) Short text that describes the structural fingerprint setup and possible reference citations. Has to be terminated by "end descr".

atom (required) The chemical species (symbol) of the central atom whose environment is captured by the setup.

env (required) A list of all atomic species that may occur in the environment of the central atom and are captured by this setup. No blank lines are allowed.

rmin (required) The minimal allowed distance between two atoms (in the distance unit used in the XSF files). This value is used by the neighbor list.

basis (required unless functions is present) Definition of a basis set for the expansion of the local atomic environment. Below, an example for the 'Chebyshev' basis type is given.

functions (required unless basis is present) Type and parameters of individual basis functions. The example below is for functions of the general type 'Behler2011', and the names of the various functions and parameters follows the original publication by Behler. No blank lines allowed.

8.2.2 Input file template using a pre-defined basis set (atom-type.stp)

```
DESCR
  short description and reference
END DESCR
```

```
ATOM <atom type>
```

```
ENV  <N>
<T_1>
<T_2>
```

```
...  
<T_N>
```

```
RMIN <R>
```

```
BASIS type=<basis type>  
<basis set parameters>
```

8.2.3 Input file template using explicit basis function definitions (atomtype.stp)

```
DESCR  
  short description and reference  
END DESCR
```

```
ATOM <atom type>
```

```
ENV <N>  
<T_1>  
<T_2>  
...  
<T_N>
```

```
RMIN <R>
```

```
FUNCTIONS type=<basis type>  
<NF>  
<parameters of function 1>  
<parameters of function 2>  
...  
<parameters of function NF>
```

8.2.4 Input file example using a Chebyshev basis set (Ti.fingerprint.stp)

The following example uses a Chebyshev basis set with a cutoff of 8.0 Å for the radial expansion (expansion order 16) and a cutoff of 6.5 Å for the angular expansion (expansion order 4).

```
DESCR
```

```

    Structural fingerprint setup for Ti in bulk TiO2.
    TiO2 reference data set:
        N. Artrith and A. Urban, Comput. Mater. Sci. 114 (2016) 135-150.
    Chebyshev descriptor:
        N. Artrith, A. Urban, and G. Ceder, Phys. Rev. B 96 (2017) 014112.
END DESCR

```

```

ATOM Ti

```

```

ENV 2
Ti
0

```

```

RMIN 0.75d0

```

```

BASIS type=Chebyshev
radial_Rc = 8.0  radial_N = 16  angular_Rc = 6.5  angular_N = 4

```

8.2.5 Input file example using explicit Behler2011 basis functions (Ti.fingerprint.stp)

```

DESCR
    Structural fingerprint setup for Ti in bulk TiO2.
    Ref.: N. Artrith and A. Urban,
        Comput. Mater. Sci. 114 (2016) 135-150.
END DESCR

```

```

ATOM Ti

```

```

ENV 2
Ti
0

```

```

RMIN 0.75d0

```

```

FUNCTIONS type=Behler2011
70
G=2 type2=O  eta=0.003214  Rs=0.0000  Rc=6.5000
G=2 type2=Ti  eta=0.003214  Rs=0.0000  Rc=6.5000

```

G=2	type2=0	eta=0.035711	Rs=0.0000	Rc=6.5000		
G=2	type2=Ti	eta=0.035711	Rs=0.0000	Rc=6.5000		
G=2	type2=0	eta=0.071421	Rs=0.0000	Rc=6.5000		
G=2	type2=Ti	eta=0.071421	Rs=0.0000	Rc=6.5000		
G=2	type2=0	eta=0.124987	Rs=0.0000	Rc=6.5000		
G=2	type2=Ti	eta=0.124987	Rs=0.0000	Rc=6.5000		
G=2	type2=0	eta=0.214264	Rs=0.0000	Rc=6.5000		
G=2	type2=Ti	eta=0.214264	Rs=0.0000	Rc=6.5000		
G=2	type2=0	eta=0.357106	Rs=0.0000	Rc=6.5000		
G=2	type2=Ti	eta=0.357106	Rs=0.0000	Rc=6.5000		
G=2	type2=0	eta=0.714213	Rs=0.0000	Rc=6.5000		
G=2	type2=Ti	eta=0.714213	Rs=0.0000	Rc=6.5000		
G=2	type2=0	eta=1.428426	Rs=0.0000	Rc=6.5000		
G=2	type2=Ti	eta=1.428426	Rs=0.0000	Rc=6.5000		
G=4	type2=0	type3=0	eta=0.000357	lambda= -1.0	zeta= 1.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.000357	lambda= -1.0	zeta= 1.0	Rc=6.5000
G=4	type2=Ti	type3=Ti	eta=0.000357	lambda= -1.0	zeta= 1.0	Rc=6.5000
G=4	type2=0	type3=0	eta=0.028569	lambda= -1.0	zeta= 1.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.028569	lambda= -1.0	zeta= 1.0	Rc=6.5000
G=4	type2=Ti	type3=Ti	eta=0.028569	lambda= -1.0	zeta= 1.0	Rc=6.5000
G=4	type2=0	type3=0	eta=0.089277	lambda= -1.0	zeta= 1.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.089277	lambda= -1.0	zeta= 1.0	Rc=6.5000
G=4	type2=Ti	type3=Ti	eta=0.089277	lambda= -1.0	zeta= 1.0	Rc=6.5000
G=4	type2=0	type3=0	eta=0.000357	lambda= 1.0	zeta= 1.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.000357	lambda= 1.0	zeta= 1.0	Rc=6.5000
G=4	type2=Ti	type3=Ti	eta=0.000357	lambda= 1.0	zeta= 1.0	Rc=6.5000
G=4	type2=0	type3=0	eta=0.028569	lambda= 1.0	zeta= 1.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.028569	lambda= 1.0	zeta= 1.0	Rc=6.5000
G=4	type2=Ti	type3=Ti	eta=0.028569	lambda= 1.0	zeta= 1.0	Rc=6.5000
G=4	type2=0	type3=0	eta=0.089277	lambda= 1.0	zeta= 1.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.089277	lambda= 1.0	zeta= 1.0	Rc=6.5000
G=4	type2=Ti	type3=Ti	eta=0.089277	lambda= 1.0	zeta= 1.0	Rc=6.5000
G=4	type2=0	type3=0	eta=0.000357	lambda= -1.0	zeta= 2.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.000357	lambda= -1.0	zeta= 2.0	Rc=6.5000
G=4	type2=Ti	type3=Ti	eta=0.000357	lambda= -1.0	zeta= 2.0	Rc=6.5000
G=4	type2=0	type3=0	eta=0.028569	lambda= -1.0	zeta= 2.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.028569	lambda= -1.0	zeta= 2.0	Rc=6.5000
G=4	type2=Ti	type3=Ti	eta=0.028569	lambda= -1.0	zeta= 2.0	Rc=6.5000
G=4	type2=0	type3=0	eta=0.089277	lambda= -1.0	zeta= 2.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.089277	lambda= -1.0	zeta= 2.0	Rc=6.5000

G=4	type2=Ti	type3=Ti	eta=0.089277	lambda= -1.0	zeta= 2.0	Rc=6.5000
G=4	type2=0	type3=0	eta=0.000357	lambda= 1.0	zeta= 2.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.000357	lambda= 1.0	zeta= 2.0	Rc=6.5000
G=4	type2=Ti	type3=Ti	eta=0.000357	lambda= 1.0	zeta= 2.0	Rc=6.5000
G=4	type2=0	type3=0	eta=0.028569	lambda= 1.0	zeta= 2.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.028569	lambda= 1.0	zeta= 2.0	Rc=6.5000
G=4	type2=Ti	type3=Ti	eta=0.028569	lambda= 1.0	zeta= 2.0	Rc=6.5000
G=4	type2=0	type3=0	eta=0.089277	lambda= 1.0	zeta= 2.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.089277	lambda= 1.0	zeta= 2.0	Rc=6.5000
G=4	type2=Ti	type3=Ti	eta=0.089277	lambda= 1.0	zeta= 2.0	Rc=6.5000
G=4	type2=0	type3=0	eta=0.000357	lambda= -1.0	zeta= 4.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.000357	lambda= -1.0	zeta= 4.0	Rc=6.5000
G=4	type2=Ti	type3=Ti	eta=0.000357	lambda= -1.0	zeta= 4.0	Rc=6.5000
G=4	type2=0	type3=0	eta=0.028569	lambda= -1.0	zeta= 4.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.028569	lambda= -1.0	zeta= 4.0	Rc=6.5000
G=4	type2=Ti	type3=Ti	eta=0.028569	lambda= -1.0	zeta= 4.0	Rc=6.5000
G=4	type2=0	type3=0	eta=0.089277	lambda= -1.0	zeta= 4.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.089277	lambda= -1.0	zeta= 4.0	Rc=6.5000
G=4	type2=Ti	type3=Ti	eta=0.089277	lambda= -1.0	zeta= 4.0	Rc=6.5000
G=4	type2=0	type3=0	eta=0.000357	lambda= 1.0	zeta= 4.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.000357	lambda= 1.0	zeta= 4.0	Rc=6.5000
G=4	type2=Ti	type3=Ti	eta=0.000357	lambda= 1.0	zeta= 4.0	Rc=6.5000
G=4	type2=0	type3=0	eta=0.028569	lambda= 1.0	zeta= 4.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.028569	lambda= 1.0	zeta= 4.0	Rc=6.5000
G=4	type2=Ti	type3=Ti	eta=0.028569	lambda= 1.0	zeta= 4.0	Rc=6.5000
G=4	type2=0	type3=0	eta=0.089277	lambda= 1.0	zeta= 4.0	Rc=6.5000
G=4	type2=0	type3=Ti	eta=0.089277	lambda= 1.0	zeta= 4.0	Rc=6.5000
G=4	type2=Ti	type3=Ti	eta=0.089277	lambda= 1.0	zeta= 4.0	Rc=6.5000

8.3 Training set generation with `generate.x`

Provided a principle input file and all required structural fingerprint setups, `generate.x` is run on the command line simply with

```
$ generate.x generate.in > generate.out
```

where `generate.in` is the principal input file, and the output will be written to `generate.out`. The code will generate a training set file that can be used for the training of ANN potentials.

The format and keywords of the principal input file are described in the

following.

8.3.1 Alphabetic list of keywords

All keywords are case insensitive and independent of the order. Blank lines and lines starting with **!**, **#**, or **%** are ignored.

debug (optional) Activate debugging mode; additional output will be generated.

files (required) Specifies number of and path to reference structures in the **ænet** XSF format. The first line following the keyword contains the number **<NF>** of structure files. Each of the **<NF>** following lines contains a file system path.

output (optional) Defines the path to the training set file that is going to be generated. The default name is "refdata.train". Note that the training set file is in a binary format and cannot be viewed by a text editor. Depending on the number of reference structures, the file can become very large (e.g., 1 GB).

setups (required) Specifies paths to structural fingerprint basis function setup files. Each of the **<NT>** lines following the keyword contains the chemical symbol **<T_{i and the path to the setup file for one species.}**

timing (optional) Activate timing; additional output files will be created.

types (required) Defines the number of atomic species, their names, and atomic energies. The first line after the keyword contains the number of different species **<NT>**; the following **<NT>** lines each contain the chemical symbol **<T_{i and atomic energy **<E_{atom-i of one species.}**}**

8.3.2 Input file template (generate.in)

OUTPUT <path/to/output/file>

TYPES

<NT>

<T_{1atom-1}

<T_{2atom-2}

```

...
<T_NT> <E_atom-NT>

SETUPS
<T_1> <path/to/setup-1>
<T_2> <path/to/setup-2>
...
<T_NT> <path/to/setup-NT>

FILES
<NF>
<path/to/file-1.xsf>
<path/to/file-2.xsf>
...
<path/to/file-NF.xsf>

```

8.3.3 Input file example (generate.in) for TiO₂

The atomic energies defined in the TYPES section is subtracted from the total energy before the potential training to reduce the fluctuations in the fitted energy (the target energy). Two different approaches towards selecting the atomic energies are shown below: In [Comput. Mater. Sci. 114 \(2016\) 135-150](#) the atomic energies are chosen to be the energies of isolated atoms. With this choice, the trained energy (total energy minus atomic energies) corresponds to the *cohesive energy*, which is reported by ænet's `predict.x` tool.

```

OUTPUT  TiO2.train

TYPES
2
O      -432.503149303  ! eV
Ti -1604.604515075   ! eV

SETUPS
O      0.fingerprint.stp
Ti Ti.fingerprint.stp

FILES

```

```

7815
./structures/0001.xsf
./structures/0002.xsf
...
./structures/7815.xsf

```

Alternatively, the atomic energies can be set to the average atomic energy of all structures in the reference data set to minimize the range of the target energy (e.g., *Phys. Rev. B* **96**, 2017, 014112). The downside of this approach is that the energy difference is no longer interpretable, i.e., it does not correspond to the cohesive energy.

OUTPUT Ti0.train

TYPES

```

2
0    -433.23448532 | eV
Ti -1626.66972707 | eV

```

SETUPS

```

0    0.fingerprint.stp
Ti Ti.fingerprint.stp

```

FILES

```

7815
./structures/0001.xsf
./structures/0002.xsf
...
./structures/7815.xsf

```

8.4 ANN potential training with train.x

ANN potential training with `train.x` requires a training set file compiled by `generate.x` (section 8.3). A number of optimization methods are implemented by `train.x`. Apart from the algorithmic differences, the methods differ in their support for parallelization and follow different learning strategies (*batch* versus *online*). For a comparison of the different training methods see the `ænet` implementation reference [1].

`train.x` expects a principal input file (named "train.in" in the example below). The tool is run from the command line with:

```
$ train.x train.in > train.out
```

where the output is written to the file `train.out`.

The format and keywords of the principal input file are described in the following.

8.4.1 Alphabetic list of keywords

All keywords are case insensitive and independent of the order. Blank lines and lines starting with `!`, `#`, or `%` are ignored.

debug (optional) Activate debugging mode; additional output files will be created.

iterations (optional) Specifies the number of training iterations/epochs (default: 10).

maxenergy (optional) Highest formation energy to include in the training set.

method (optional) Specifies the training method/algorithm to be used for the weight optimization. The line following the keyword contains as first item the name of the method (e.g., `bfgs`, `online_gd`, `lm`) and as further items the parameters of the method (if applicable). The default method is `bfgs`.

networks (required) Defines the architectures and specifies files for all ANNs. Each of the `<NT>` (= number of types) lines following the keyword contains the chemical symbol `<T_i>` of the i -th atomic species in the training set, the path to the ANN output file (binary), and the architecture of the hidden network layers. The latter is defined by the number of hidden layers followed by the number of nodes and the activation function separated by a colon (see example below for two hidden layers of 5 nodes each and the hyperbolic tangent activation).

save_energies (optional) Activate output of the final energies of all training and testing structures. The resulting output files can be used to visualize the quality of the ANN fit and to identify structures that are not well represented. One file per process will be generated, containing

only the energies of all structures handled by the process. The files can simply be concatenated.

testpercent (optional) Specifies the percentage of reference structures to be used as independent testing set (default: 10%).

timing (optional) Activate timing; additional output files will be created.

trainingset (required) Defines the name/path to the binary training set file (output of generate.x, e.g., "refdata.train").

8.4.2 Training methods

The training method is specified with the **method** keyword followed by the identifier of the method and its parameters. Currently, **train.x** offers three different optimization methods: online gradient descent, the limited-memory BFGS algorithm and the Levenberg-Marquardt method.

1. Online gradient descent (**online_gd**)

Gradient descent is implemented as *online* learning method which currently prevents efficient parallelization. The method is selected with the identifier **online_gd** and has two parameters, the *learning rate* (**gamma**) that is a measure of the stepsize per iteration, and the *momentum parameter* (**alpha**) that controls fluctuations.

An example definition with reasonable parameters is:

```
METHOD
online_gd gamma=3.0d-2 alpha=0.05d0
```

2. Limited-Memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS) method

The L-BFGS method is implemented as *batch* training method, which enables efficient parallelization of the error function evaluation. The method is selected with the identifier **bfgs** and does not currently offer any adjustable parameters:

```
METHOD
bfgs
```

3. Levenberg-Marquardt method

The Levenberg-Marquardt method that is presently only available in serial is selected with the identifier `lm`. The method supports a number of parameters: `batchsize` sets the number of training points that are used to evaluate the error function at a time. This *batch size* determines the computational requirements of the method, but should be chosen as large as possible to guarantee convergence. The `learnrate` is the initial value of the learning rate (see online gradient descent). The parameter `iter` determines the number of iterations per optimization step used to adjust the learning rate, and the factor used for this adjustment is defined with `adjust`. Finally, a convergence threshold for the error function can be specified with `conv`.

Example of reasonable parameters

```
METHOD
lm batchsize=5000 learnrate=0.1d0 iter=3 conv=0.001 adjust=5.0
```

8.4.3 Input file template (train.in)

```
TRAININGSET <path/to/data/file>
TESTPERCENT <percentage>
ITERATIONS <NI>
MAXENERGY <emax e.g. -0.05 eV>
SAVE_ENERGIES
```

```
METHOD
<method name> <parameters>
```

```
# Examples
#
# (1) online steepest descent
# METHOD
# online_gd gamma=5.0d-7 alpha=0.25d0
# (2) BFGS
# METHOD
# bfgs
# (3) Levenberg-Marquardt
# METHOD
# lm batchsize=1000 learnrate=0.1 iter=1 conv=0.001 adjust=10.0
```

```

NETWORKS
# atom    network          hidden
# types   file-name        layers  nodes:activation
<T_1>    <path/to/net-1>     2       5:tanh  5:tanh
<T_2>    <path/to/net-2>     2       5:tanh  5:tanh
...
<T_NT>   <path/to/net-NT>    2       5:tanh  5:tanh

# Example using different activation functions:
# For details see Eq. (1) in:
# N. Artrith and A. Urban, Comput. Mater. Sci. 114 (2016) 135-150.
#
# <T_1>    <path/to/net-1>     2       5:linear 5:linear
# <T_2>    <path/to/net-2>     2       5:linear 5:linear

# <T_1>    <path/to/net-1>     2       5:tanh   5:tanh
# <T_2>    <path/to/net-2>     2       5:tanh   5:tanh

# <T_1>    <path/to/net-1>     2       5:sigmoid 5:sigmoid
# <T_2>    <path/to/net-2>     2       5:twist  5:twist

```

8.4.4 Example input file (train.in)

```

TRAININGSET TiO2.train
TESTPERCENT 10
ITERATIONS 500

TIMING

METHOD
lm batchsize=5000 learnrate=0.1d0 iter=3 conv=0.001 adjust=5.0

NETWORKS
! atom    network          hidden
! types   file-name        layers  nodes:activation
0         0.10t-10t.ann     2      10:twist 10:twist
Ti        Ti.10t-10t.ann    2      10:twist 10:twist

```

8.5 Restarting training from existing ANN potential

During the training process, **ænet** creates the restart files `train.restart` and `train.rngstate` that contain all information needed to continue the training where it was interrupted. These files will automatically be used when present. If `train.x` was terminated it might additionally be necessary to copy the most recent ANN weights stored in the final `*.ann-XXX` files (where `XXX` is the number of the final training epoch) to corresponding files without epoch number (i.e., simply `*.ann`).

If no restart is desired, the file `train.restart` has to be deleted.

Note: Currently, restarting is only implemented for the BFGS training method.

9 Using ANN potentials for atomistic simulations

It is not the aim of the **ænet** package to compete with well-established and feature-rich software for molecular dynamics and Monte-Carlo simulations, such as [LAMMPS](#), [DL_POLY](#), [TINKER](#), or [ASE](#). Instead, **ænet** provides a library with C and Fortran APIs, **ænetLib**, that can be used to extend existing software by the capability to evaluate ANN potentials constructed with **ænet**'s `train.x`. Note that software developed in many other programming languages (e.g., C++, Python, and Java) can interface with C libraries and, hence, is compatible with **ænetLib**.

A documentation of the **ænetLib** APIs will be included in a future version of this manual. For the moment, **ænet** provides two reference implementations for the evaluation of structural energies and forces by linking against **ænetLib**: `predict.x` is written in Fortran and directly uses the Fortran API, and `ænet-predict.py`, which implements an [ASE](#) *calculator* in Python. In addition, an example Python script for performing simple molecular dynamics simulations with ASE, `ænet-md.py`, is included in the **ænet** package.

9.1 Prediction of structural energies and atomic forces with `predict.x`

`predict.x` expects a principal input file (named "predict.in" in the example below) and one or more atomic structure files in the XSF format. The path(s) to the structure files may either be specified in the input file for

batch processing, or directly on the command line. The tool is run from the command line with:

```
$ predict.x predict.in [<structure1.xsf> ...]
```

All output will be written to standard out.

The format and keywords of the principal input file are described in the following.

9.1.1 Alphabetic list of keywords

All keywords are case insensitive and independent of the order. Blank lines and lines starting with **!**, **#**, or **%** are ignored.

debug (optional) Activate debugging mode; additional output files will be created.

files (optional) Specifies a list of paths to input structures. This keyword may be used for batch processing of a larger number of structures. The line following the keyword contains the number of input files **<NF>**, and each of the following **<NF>** lines contains a single file system path. Alternatively, a single input structure may be passed to **predict.x** as command line argument. The command line takes precedence over the list specified with the "files" keyword.

forces (optional) Activates evaluation of the atomic forces. Forces are also calculated, when the "relax" keyword is present.

networks (required) Specifies the ANN potential files for each chemical species. On each of the **<NT>** lines following the keyword a chemical species **<T_i>** and the path to its corresponding ANN file is given.

relax (optional) Activate structural relaxation; this will automatically also activate the calculation of the atomic forces. On the line following the **relax** keyword, several options can be specified. See the example below.

timing (optional) Activate timing; additional output files will be created.

types (required) Specifies the number of different atomic species that may occur in structures and their chemical symbols. The first line following the keyword specifies the number **<NT>** of different atom types; the following lines each contain one chemical symbol **<T_i>**.

9.1.2 Input file template (predict.in)

```
TYPES
<NT>
<T_1>
<T_2>
...
<T_NT>

NETWORKS
<T_1> <path/to/NN-1>
<T_2> <path/to/NN-2>
...
<T_NT> <path/to/NN-NT>

FORCES

# or optimize coordinates:
#
# RELAX
# method=bfgs  F_conv=1.0d-2  E_conv=1.0d-6  steps=99
#
#   method: optimization method (currently only BFGS)
#   F_conv: convergence thershold for the forces
#   E_conv: convergence threshold for the energy
#   steps:  max. number of iterations

FILES
<NF>
<path/to/structure-1.xsf>
<path/to/structure-2.xsf>
...
<path/to/structure-NF.xsf>
```

9.1.3 Input file example (predict.in) for TiO₂

```
TYPES
2
Ti
```

0

NETWORKS

Ti Ti.10tw-10tw.ann
O O.10tw-10tw.ann

FORCES

FILES

10

structure0001.xsf
structure0002.xsf
structure0003.xsf
structure0004.xsf
structure0005.xsf
structure0006.xsf
structure0007.xsf
structure0008.xsf
structure0009.xsf
structure0010.xsf

9.2 ASE Interface: `aenet-predict.py` and `aenet-md.py`

The *Atomic Simulation Environment* (ASE) is a Python framework for atomistic simulations and for the manipulation of atomic structures. ASE provides a simple API, *calculators*, for interfacing with third-party software for the evaluation of structural energies and atomic forces. The **aenet** package includes an implementation of an ASE calculator linked to **aenetLib**. The script `aenet-predict.py` uses this calculator to essentially replicate the features of `predict.x` (see above), and `aenet-md.py` provides simple molecular dynamics capabilities.

The input files for both Python scripts use the [JSON](#) format and are compatible. Any structure format supported by [ASE](#) can be used as input, however, as of writing, the support of the XSF structure format in ASE is incomplete and other formats (e.g., VASP's POSCAR format, FHI-aims geometry.in format, XYZ, etc.) are recommended.

9.2.1 Alphabetic list of keywords

The input files of `aenet-predict.py` and `aenet-md.py` both use the [JSON](#) format. Keywords that are specific to one tool are ignored by the other.

potentials (required) Specifies the ANN potentials for all atomic species.

structure_file (MD only) Path to the file with the initial structure. Every structure format that is understood by ASE can be used.

trajectory_file (MD only) Path to the trajectory file (in ASE's format) to be generated during the MD simulation.

temperature (MD only) Temperatur for MD simulations in the canonical ensemble.

md_steps (MD only) Number of MD steps.

print_steps (MD only) Number of MD steps between writing output.

time_step (MD only) MD time step in femtoseconds.

9.2.2 Input file template (input.json)

```
{
  "potentials" : {
    <T1> : <potential1>,
    <T2> : <potential2>,
    ...
  },
  "structure_file" : <initial-structure>,
  "trajectory_file" : <output-file>,
  "temperature" : <T>,
  "md_steps" : <N_MD>,
  "time_step" : <dt>,
  "print_steps" : <N_print>
}
```

9.2.3 Input file example (input.json)

```
{
```

```

    "potentials" : {
        "Ti" : "Ti.10t-10t.ann",
        "O"  : "O.10t-10t.ann"
    },
    "structure_file" : "input.vasp",
    "trajectory_file" : "md.traj",
    "temperature" : 300.0,
    "md_steps" : 100,
    "time_step" : 1.0,
    "print_steps" : 1
}

```

10 Acknowledgment

This work used the [Extreme Science and Engineering Discovery Environment \(XSEDE\)](#), which is supported by National Science Foundation grant number ACI-1053575.

11 Questions?

If you run into problems with **ænet** or if you have a general question, please contact Dr. Nongnuch Artrith (nartrith@atomistic.net).

12 Bibliography

- [1] N. Artrith and A. Urban, *Comput. Mater. Sci.* **114** (2016) 135-150.
- [2] N. Artrith, A. Urban, and G. Ceder, *Phys. Rev. B* **96** (2017) 014112.
- [3] J. Behler and M. Parrinello, *Phys. Rev. Lett.* **98** (2007) 146401.
- [4] J. Behler, *J. Chem. Phys.* **134** (2011) 074106.
- [5] A. P. Bartók, M. C. Payne, R. Kondor, and G. Csányi, *Phys. Rev. Lett.* **104** (2010) 136403.
- [6] R. H. Byrd, P. Lu and J. Nocedal, *SIAM J. Sci. Stat. Comp.* **16** (1995) 1190-1208.
- [7] K. Levenberg, *Q. Appl. Math.* **2** (1944) 164–168.

- [8] D. W. Marquardt, *SIAM J. Appl. Math.* **11** (1963) 431–441.